

Device Simulation Layer

Simulated Office (Wokwi) ESP32 + 15 Devices

Rust Tokio Simulator Background Task (30-60s)

Writes state history

Rust Backend (Single Source of Truth)

Proactive 60s Alert Loop

Axum REST Router

/diagram (Tectonic Compile)

Polling / REST

React/Vite Dashboard

Uploads PDF

Presigned URL Download

AWS S3 (Files)

Uploads CSV

/devices (Toggle/Status)

/usage (Live Power & kWh)

/alerts (Anomaly Engine)

/reports (CSV Exporter)

Direct Read/Write

Shared AppState (RwLock)

Humanize Output

rig LLM Pipeline(Gemini / OpenAI)

Discord API

Serenity Discord Bot Background Task

In-Memory Fallback Cache(Graceful Degradation)

PostgreSQL Primary DB

Redis Rate Limit & Cache

